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RESEARCH

Contrasting binding architectures underlying the asymmetric M327I resistance of menin inhibitors

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Abstract

Context The interaction between menin and mixed lineage leukemia fusion proteins represents a critical therapeutic target. The clinical inhibitors revumenib and ziftomenib display divergent resistance profiles against the M327I mutation, which frequently compromises revumenib efficacy while ziftomenib maintains potency. Molecular dynamics simulations of a complete 2×2 comparison matrix (WT/M327I × Revumenib/Ziftomenib, three independent 500 ns replicates per system) suggest that this asymmetry arises from contrasting binding architectures. Revumenib depends on an intrinsically fragile Trp346 hydrogen bond anchor that is disrupted in multiple independent trajectories, including both M327I and WT backgrounds. Ziftomenib distributes binding energy across a redundant hydrophobic network with stable Trp346 engagement across all replicates and minimal sensitivity to position-327 perturbation. Energy decomposition shows the substantial Ile327 contribution originates from forced close-range contact, whereas the minimal Met327 contribution in ziftomenib complexes aligns with distributed network architecture. The contrasting architectures may inform the design of therapeutics with improved mutational tolerance.

Methods Twelve independent 500 ns molecular dynamics simulations (four systems × three replicates) were performed using OpenMM with AMBER ff14SB for protein and OpenFF Sage for ligands. Systems were solvated with TIP3P water. Binding free energies and per-residue contributions were estimated via MM/GBSA with the GB OBC2 implicit solvent model. Principal component analysis and dynamic cross-correlation matrices characterized conformational sampling.

Keywords Menin-KMT2A inhibitor · Molecular dynamics simulation · Drug resistance mechanism · MM/GBSA energy decomposition · Binding architecture asymmetry

Introduction

Rearrangements of the mixed lineage leukemia gene KMT2A (formerly MLL) define a distinct subset of acute leukemias that account for 5 to 10% of cases in both children and adults and are associated with poor clinical outcomes [1]. The leukemogenic activity of KMT2A fusion proteins depends on their direct interaction with menin, a chromatin-associated scaffold protein encoded by the MEN1 gene [2]. Menin recruits KMT2A fusion proteins to the promoters of HOXA cluster and MEIS1 genes, sustaining the aberrant transcriptional program that maintains the leukemic state [3, 4]. Genetic ablation of menin or pharmacologic disruption of the menin-KMT2A protein-protein interaction has been shown to reverse this oncogenic transcription and induce differentiation and apoptosis of leukemia cells [2, 3], supporting the menin-KMT2A interface as a therapeutic target.

Two small-molecule menin inhibitors with distinct chemical scaffolds have advanced to late-stage clinical development. Revumenib (SNDX-5613) and ziftomenib (KO-539) differ substantially in their core scaffolds and binding-site engagement geometry. Revumenib is built on a bicyclic scaffold comprising a piperidine ring fused to a triazole, with a cyclohexyl substituent extending from the piperidine nitrogen and a para-substituted phenylsulfonamide projecting into the binding pocket [5]. Ziftomenib features a structurally distinct core consisting of a pyrrolidine-linked thiazole with

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an extended tail carrying a trifluoromethyl-substituted aromatic ring. This extended geometry projects further into the binding pocket and allows engagement with a broader set of polar and hydrophobic contacts, including residues not accessed by the Revumenib pharmacophore [5, 6]. The two inhibitors have fundamentally different binding architectures that may underlie their divergent resistance profiles. Revumenib demonstrated a complete remission or complete remission with partial hematologic recovery rate of 22.8 percent in the phase 2 AUGMENT-101 trial and received accelerated approval from the United States Food and Drug Administration in 2024 for relapsed or refractory KMT2A-rearranged acute leukemia [7, 8]. Ziftomenib showed comparable efficacy in NPM1-mutated acute myeloid leukemia in the KOMET-001 trial and is currently under evaluation in phase 3 studies [9, 10]. Despite their shared target, the two inhibitors exhibit markedly different resistance profiles. Acquired mutations in MEN1, most frequently M327I (615-amino acid isoform numbering), were identified in approximately 40 percent of revumenib-treated patients who developed clinical resistance [5]. In contrast, base-editor screening and functional validation showed that mutational resistance profiles differ substantially between menin inhibitors, with some substitutions conferring pan-class effects while others display agent-selective sensitivity [6]. Structural analysis of the menin(M327I)-revumenib co-crystal structure (Protein Data Bank entry 8E90, resolution 1.85 Å) suggested that the branched side chain of isoleucine at position 327 projects toward the cyclohexyl ring of revumenib, generating steric conflict and disrupting the hydrogen bond with Trp346 [5]. A crystallographic snapshot captures only a single energy-minimum conformation. Whether the steric conflict causes immediate ligand rejection or progressive weakening of the anchor interaction remains unclear. Whether the identical substitution exerts asymmetric effects on two inhibitors with distinct chemical scaffolds also remains unclear. Molecular dynamics simulations can address these questions by tracking the temporal evolution of each protein-ligand complex under physiological conditions.

Beginning from the menin(M327I)-revumenib co-crystal structure (Protein Data Bank entry 8E90) and the menin(wild-type)-ziftomenib co-crystal structure (Protein Data Bank entry 8VA6, resolution 1.57 Å), we used in silico point mutations to construct the two reciprocal complexes and assembled a complete two-by-two matrix of wild-type and M327I backgrounds for each inhibitor. Each of the four complexes was sampled with three independent 500 ns all-atom replicates, 12 trajectories in total, and analyzed across global dynamic stability, anchor residue behavior, hydrogen bond network persistence, conformational sampling and correlated motions, and binding free energy decomposition. The simulations suggest that the two inhibitors employ contrasting binding strategies. Revumenib couples to an intrinsically fragile Trp346 anchor that is disrupted in multiple trajectories, including both M327I and WT backgrounds. Energy decomposition shows the substantial Ile327 contribution originates from forced close-range contact. Ziftomenib distributes binding energy across a redundant hydrophobic network with stable Trp346 engagement and minimal sensitivity to position 327. These observations may contribute to an atomic-level understanding of the asymmetric M327I resistance profile and inform the design of next-generation menin inhibitors.

Results

Global dynamic stability and conformational plasticity

All 12 trajectories reached backbone equilibration within the first 100 ns, after which the single-chain Cα RMSD fluctuates about a stable mean with standard deviation $\leq$ 0.34 Å (Fig. 1a; Figs. S1 and S2). Post-equilibration values are 2.15±0.28 Å for M327I–Revumenib, 2.33±0.30 Å for WT–Revumenib, 2.47±0.26 Å for WT–Ziftomenib and 3.13±0.36 Å for M327I–Ziftomenib (mean±SD, n=3, 100–500 ns). The radius of gyration remains tightly clustered at 24.98–25.29 Å across all replicates (Fig. 1b; per-trajectory metrics in Table S1), providing no evidence for scaffold expansion or compaction. The first 100 ns are treated as equilibration and excluded from binding free energy analyses. The anchor weakening at approximately 380 ns occurs after backbone equilibration.

Ligand RMSD separates the two inhibitors more clearly than any protein-level metric. Ziftomenib remains positionally stable in both backgrounds (1.08±0.20 Å in the wild type and 1.16±0.08 Å in M327I), whereas Revumenib is more variable (1.30±0.31 and 1.74±0.44 Å), with the largest replicate values arising in the trajectories that lose the Trp346 anchor (Fig. 2). Construct differences (8E90 dimer vs 8VA6 monomer) preclude cross-inhibitor RMSD comparison; within-ligand comparisons isolate substitution effects.

Per-residue backbone fluctuation was computed for all 12 trajectories on clinical numbering derived by residue-level alignment to the source crystal structures (Fig. 1c). Within each trajectory, the binding pocket is more ordered than the remainder of the chain. The ratio of mean pocket RMSF to mean non-pocket RMSF is below unity in every one of the 12 trajectories, ranging from 0.66 to 0.83 with a mean of 0.75±0.05. Because this ratio is formed within a single trajectory, it is unaffected by the replicate-to-replicate variation discussed below. Pocket order is therefore a property shared by all four complexes rather than a feature that distinguishes them.

RMSF profiles show replicate-to-replicate variation comparable to between-complex differences (0.25±0.09 vs

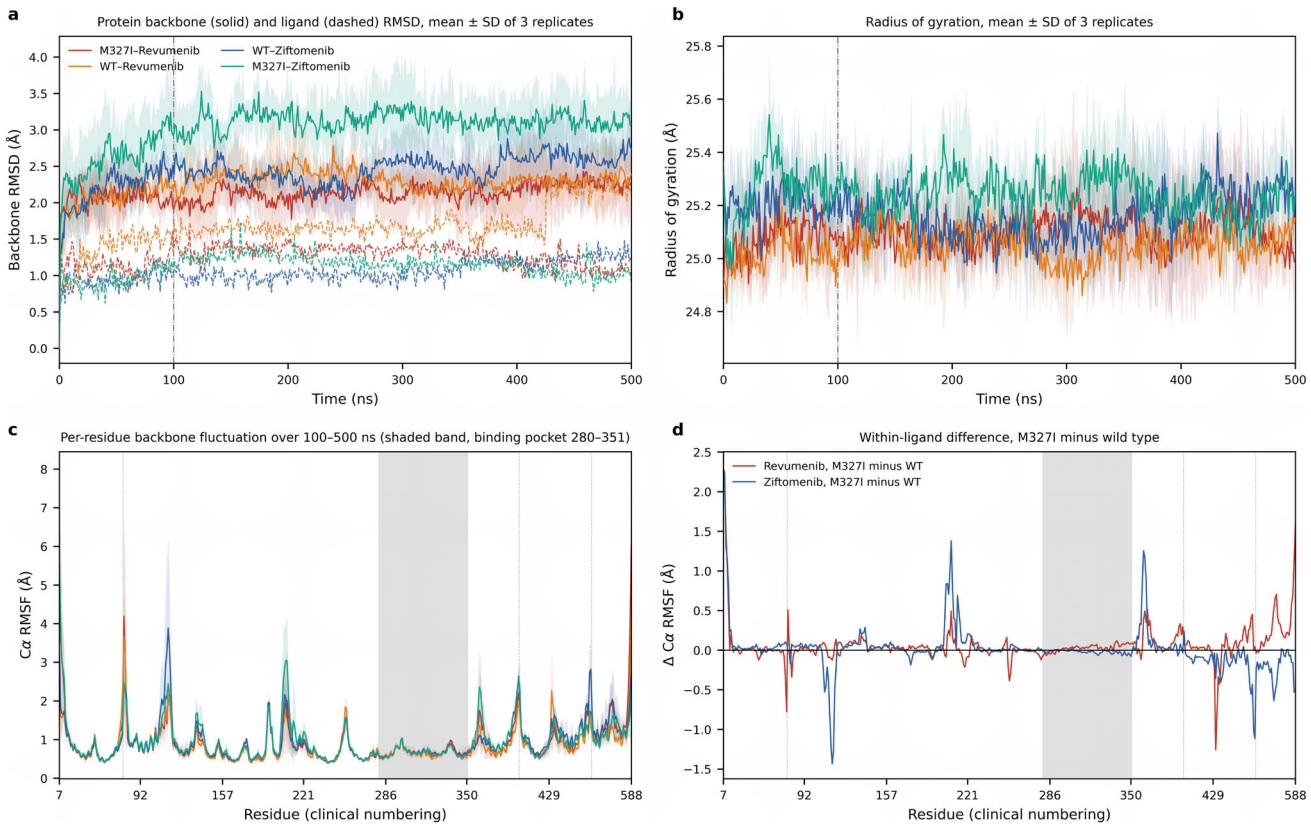


Fig. 1 Global dynamics of the four Menin complexes over twelve 500 ns trajectories. All panels show the mean of three independent replicates per complex with shading indicating the replicate standard deviation. **a** Backbone Cα RMSD (solid lines) and ligand RMSD (dashed lines) versus simulation time. **b** Radius of gyration. **c** Per-residue Cα RMSF over the 100–500 ns window. Residue numbering follows the 615-residue isoform; dotted vertical lines mark crystallographic chain-segment boundaries. Grey band: binding pocket (residues 280–351). **d** Within-inhibitor RMSF difference (M327I minus wild type) for each inhibitor

0.26±0.08 Å for Revumenib systems), precluding group-level distinctions at 500 ns per replicate (Figs. S6c and 1d).

Anchor residue dynamics and binding interaction patterns

Trp346 NE1 donor-acceptor distance shows an asymmetry centered on position 327 and its relationship to the behavior of the Trp346 anchor (Fig. 3a–d).

In the M327I–Revumenib complex, the distance evolution shows three phases (Fig. 3a). Large initial fluctuations (0–110 ns, spikes reaching 6–8 Å) likely reflect equilibration from crystallographic geometry to a fully solvated state. Even at this stage, the Ile327 side chain maintains sub-4.0 Å contact with the ligand from the first nanoseconds of simulation across all three replicates, consistent with persistent structural proximity of the isoleucine branched side chain to the ligand (supplementary contact profiles).

Between approximately 110 and 380 ns, the Trp346 distance partially recovers to the 3–4 Å range, suggesting a metastable binding state in which Revumenib retains marginal Trp346 NE1 hydrogen bond contact with its primary anchor. This state appears unstable.

One simulation undergoes complete disengagement throughout the production period; a second shows progressive loss after approximately 380 ns; the third maintains contact in 82% of frames. The WT–Revumenib simulations suggest the same fragility. One of three simulations undergoes complete disengagement (0% occupancy); the other two maintain stable contact throughout (93% and 92%). Trp346 engagement therefore appears intrinsically fragile for Revumenib binding independent of the M327I mutation. The mutation appears to exacerbate an existing instability rather than create a novel failure mode. Where the anchor disengaged, the ligand receded from the Ile327 contact zone (Fig. 4a), consistent with a local repositioning of the drug within the pocket rather than a mutation-specific steric ejection.

The WT–Ziftomenib and M327I–Ziftomenib complexes present a contrasting picture (Fig. 3c, d). All three WT–Ziftomenib simulations maintain Trp346 NE1 contact below 3.5 Å in more than 90% of frames (93%, 91%, and 93%), and all three M327I–Ziftomenib simulations remain stably

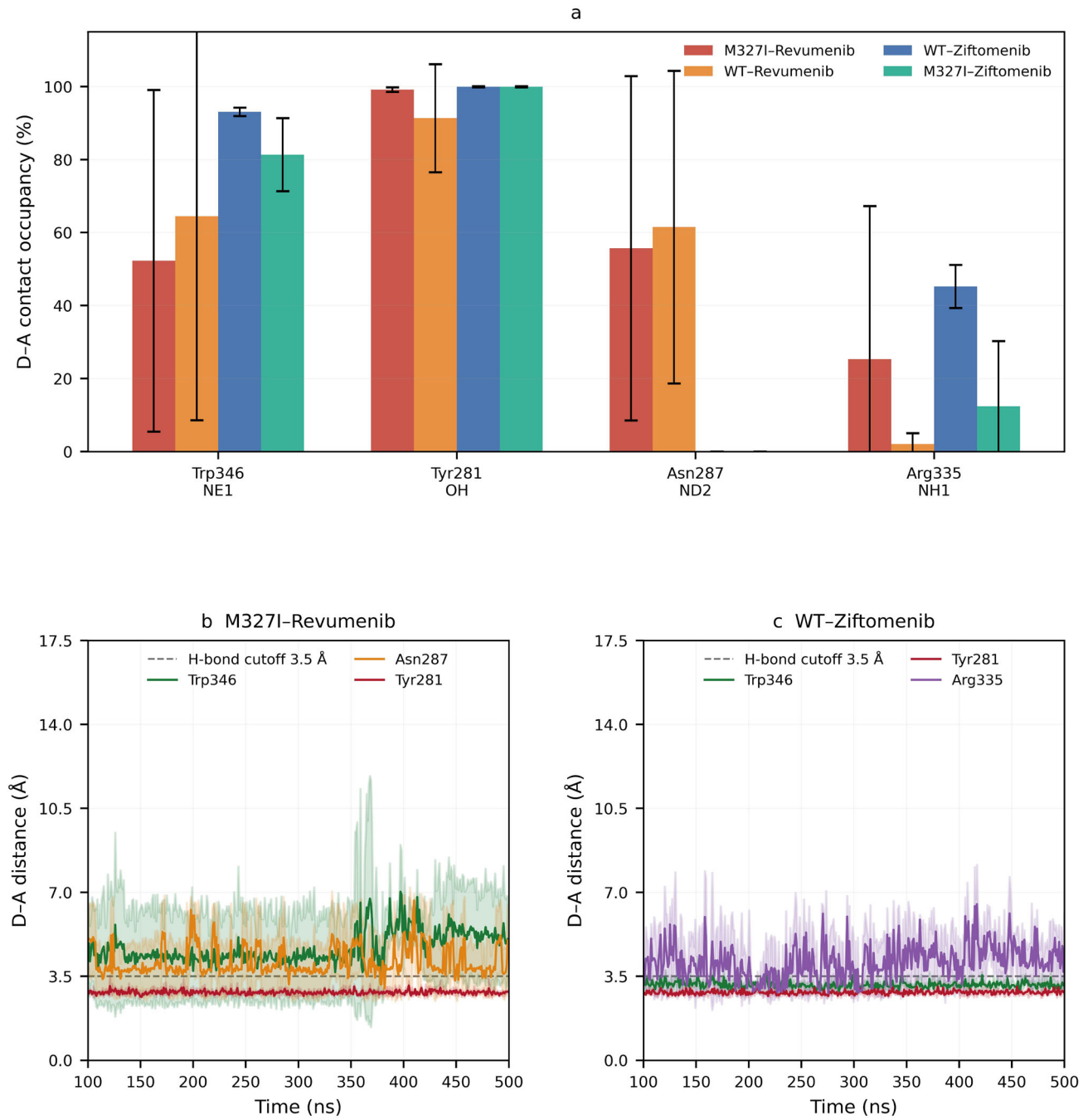


Fig. 2 Hydrogen-bond network occupancy and dynamics across all four Menin–inhibitor systems. a D–A contact occupancy (D–A < 3.5 Å, Baker–Hubbard criteria, 100–500 ns) for four key donor–acceptor atoms across all four complexes. Bars show the mean of three independent replicates; error bars show replicate standard deviation. **b** Mean ± SD D–A distance time series for M327I–Revumenib (100–500 ns, $n = 3$). Green: Trp346 NE1; orange: Asn287 ND2; red: Tyr281 OH. Shading indicates replicate standard deviation. Dashed line: 3.5 Å threshold. **c** Equivalent profiles for WT–Ziftomenib. Green: Trp346 NE1; red: Tyr281 OH; purple: Arg335 NH1. Profiles for WT–Revumenib and M327I–Ziftomenib are provided in Fig. S4

engaged (90%, 76%, and 71%). Met327 maintains modest contact as one member of a broader hydrophobic ensemble in the WT–Ziftomenib complex, with Tyr281, Tyr324, Tyr328, and Met283 each remaining consistently below 5 Å, and no single residue appearing to bear disproportionate anchoring responsibility. Structural snapshots at 250 ns illustrating the pre-transition geometry in M327I–Revumenib and the distributed hydrophobic network in WT–Ziftomenib are provided in the supplementary material.

Pocket-aligned ligand RMSD separates the two inhibitors (computed after superposing binding-site Cα atoms to isolate ligand motion) (Fig. 4a). All six Ziftomenib trajectories hold a mean below 1.7 Å (1.23–1.66 Å), whereas the M327I–Revumenib trajectories average 1.96–2.45 Å and spend only 9 to 41 percent of frames below 2 Å. One WT–Revumenib replicate showed substantial ligand reorientation (pre-post RMSD 4.96 Å), consistent with the view that Revumenib binding may be intrinsically labile even in a wild-type background. Ziftomenib trajectories maintained consistently low RMSD across all replicates, consistent with a more stable binding mode. The ligand–pocket center-of-mass distance showed the same contrast (Fig. 4b), with the Ziftomenib complexes holding a stable separation near 6.7–6.8 Å (mean, n=3) while the Revumenib complexes sampled larger and more variable distances (M327I–Revumenib 7.3 Å; WT–Revumenib 8.1 Å, the latter inflated by the reorienting replicate).

Pairwise correlation of ligand–residue minimum distances was computed for the residue triad at clinical positions 346, 330 and 372, comprising the Trp346 anchor itself and the two pocket residues whose ligand distances track it most closely. In M327I–Revumenib, the mean absolute internal correlation of the triad is 0.61±0.06, against a background of 0.10±0.03 taken over non-adjacent pocket residue pairs in the same trajectories, a factor of 5.9 (Fig. 4d). The triad also exceeds its own background in the other three systems, by factors of 2.7 for WT–Revumenib, 2.8 for WT–Ziftomenib, and 3.7 for M327I–Ziftomenib, so correlated motion of this triad appears to be a feature shared by all four complexes and the difference between systems is one of degree. The largest within-ligand difference is in the Revumenib background, where the triad correlation is 0.61±0.06 with Ile327 against 0.26±0.19 with Met327. Two qualifications apply. The triad correlation is undefined in one replicate per Revumenib group: in M327I–Revumenib, the ligand was retained but stable Trp346 contact was never established (Trp346 occupancy 0%, group entry therefore excluded); in WT–Revumenib, the ligand departed from the binding pocket during the early phase. Each Revumenib group therefore contributes two rather than three values, and the surviving trajectories are by construction those that retained the ligand. The two WT–Revumenib values also differ by a factor of three (0.12 and 0.39). The elevated triad coupling in M327I–Revumenib is therefore a candidate mutation-associated feature rather than an established one at this sampling depth. This coupling is further corroborated by the sulfonamide hydrogen bond network: Bond A (Trp346 NE1 ⋯ sulfonamide oxygen) and Bond B (sulfonamide nitrogen ⋯ residue 327 backbone carbonyl) showed near-identical fractional occupancy within each replicate, switching synchronously with the anchor state, consistent with both bonds bridging a single helix-turn-helix structural module (Figs. 4i, j and S5). Across the six Revumenib replicates, the two occupancies differ by 1.0±1.2 percentage points, and the two bonds occupy the same state in 98.7±1.2% of frames (Table S3).

Quantification of the helix centroid separation (Fig. S5) yields mean values of 8.81±0.09, 8.96±0.06, 8.85±0.01, and 9.10±0.44 Å for M327I–Revumenib, WT–Revumenib, WT–Ziftomenib, and M327I–Ziftomenib respectively, so the helix separation itself does not distinguish the four systems. In the two M327I–Revumenib replicates that undergo anchor disengagement (replicates 1 and 3), the helix separation and anchor distance time series are positively correlated (Pearson $r = 0.43$ and $r = 0.31$) against a background of $r = 0.15 \pm 0.09$ (range -0.05 to 0.25) across the remaining ten replicates, consistent with the helix-opening and anchor-loss coupling described by Perner et al. [5]. M327I–Ziftomenib replicate 1 shows the largest helix separation of all 12 trajectories (mean 9.6 Å) yet maintains stable Trp346 anchor contact throughout, consistent with Ziftomenib binding not requiring helix-turn-helix bridging.

Side-chain conformational flexibility at position 327, quantified as the circular standard deviation of the χ_2 dihedral angle across the equilibrated trajectory window, differed between Met and Ile residues, though the two ranges are not fully separated. Met327 in the WT–Ziftomenib and WT–Revumenib systems showed a χ_2 circular standard deviation of 60–110°. Ile327 spans 17–62° across the six M327I replicates, 17–40° in the Revumenib background and 25–62° in the Ziftomenib background, so the highest Ile327 value overlaps the lowest Met327 value (Fig. 4f). The separation is therefore clear in the Revumenib background and marginal in the Ziftomenib background. This difference reflects in part the intrinsic conformational preferences of a branched β-carbon side chain relative to a linear sulfur-containing one, and χ_2 describes different atom quadruplets in the two residues, so the values are not strictly comparable. The reduced flexibility of Ile327 in the Revumenib complexes may limit its capacity to accommodate fluctuations in the anchor-proximal contact module.

Hydration of the anchor site was re-examined at 0.1 ns resolution (Fig. 4e). While the Trp346 NE1 contact is intact, the site is dry (mean 0.00–0.01 water oxygens within 3.5 Å of NE1) over more than 600 ns, suggesting solvent insertion does not occur into the intact contact. Two trajectories, one M327I–Revumenib and one WT–Revumenib replicate, contribute no engaged frames because the ligand had left the pocket before the analysis window opened. Water appears only once the contact is lost, rising to approximately one oxygen and not increasing further, consistent with a single water occupying the hydrogen-bonding position vacated by the ligand rather than with solvent driving the ligand away. Even in the WT–Revumenib replicate in which the ligand recedes beyond 15 Å, NE1 hydration reaches only 0.69, so the site does not become fully solvent exposed. Hydration

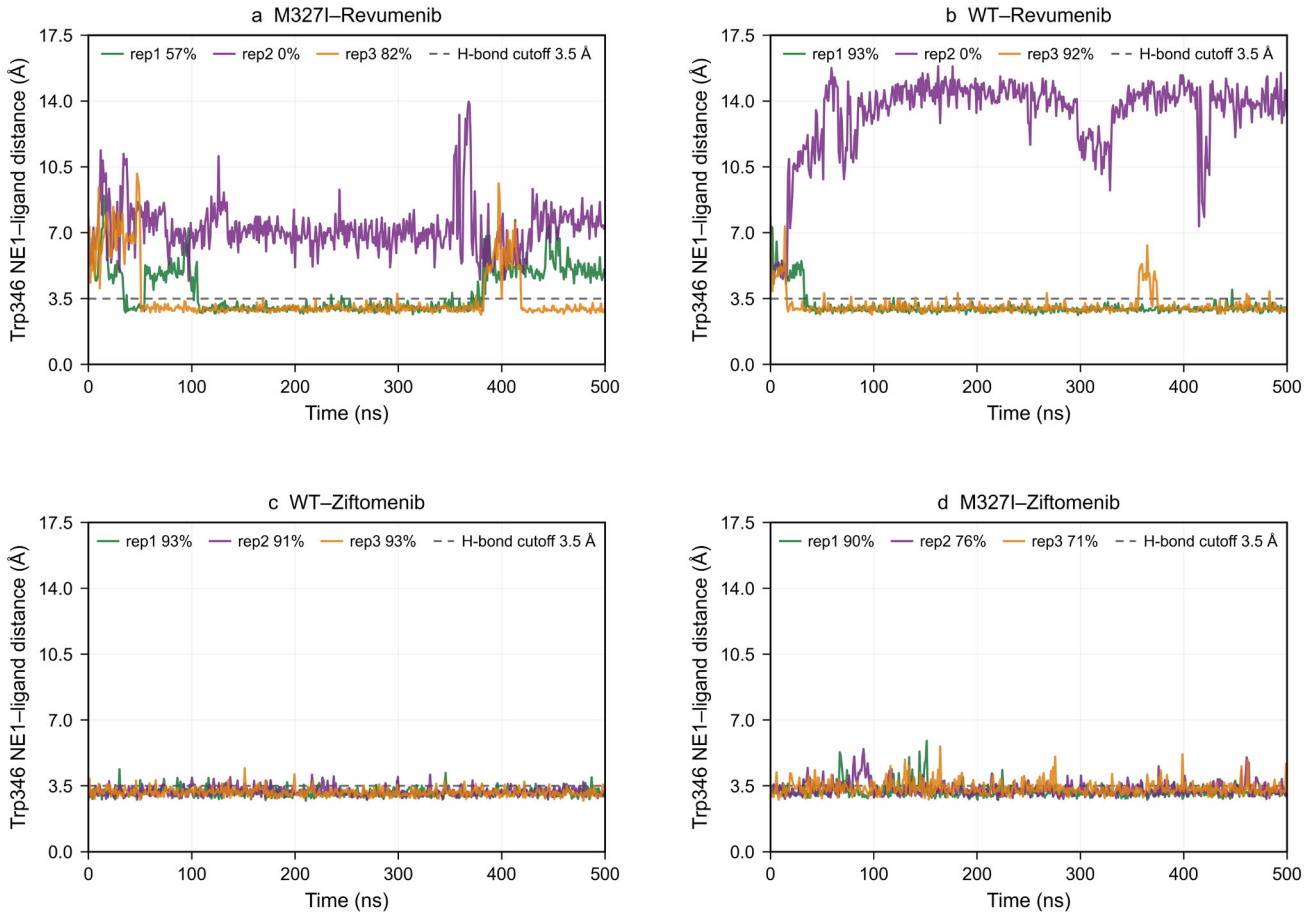


Fig. 3 Trp346 NE1 hydrogen bond distance to the ligand across all four Menin–inhibitor systems. Each panel shows the distance from the Trp346 NE1 nitrogen (the indole ring hydrogen bond donor) to the nearest ligand heavy atom, sampled at every tenth frame across 500 ns of production simulation. Three independent simulations per system are shown as individual traces; the dashed line marks the 3.5 Å donor–acceptor distance criterion for hydrogen bond engagement. Occupancy denotes the fraction of frames in which this distance falls below 3.5 Å and is reported for each simulation in the panel legend. **a** M327I–Revumenib. **b** WT–Revumenib. **c** WT–Ziftomenib. **d** M327I–Ziftomenib. Structural snapshots and full pocket residue contact profiles are provided in the supplementary material

tracks the anchor distance alone, and the four systems agree wherever their distance ranges overlap, so these trajectories give no evidence for a ligand-specific hydration effect at matched anchor geometry. The two inhibitors differ in how far the anchor travels rather than in how much water enters at a given distance. The Ziftomenib anchor spends less than 0.4% of frames beyond 5 Å in each of six replicates, so it has no appreciably populated hydrated range. The relative timing of solvent entry and contact rupture cannot be resolved here, because only four sustained disengagement events occur across the 12 trajectories and the water count in the nanosecond preceding them is inconsistent between them, at 0.00, 0.50, 1.00 and 1.00 oxygens. Clustering of frames before and after the 380 ns boundary yielded pre- and post-transition representative structures for each trajectory; the main-text figure shows replicate 1 of M327I–Revumenib and WT–Ziftomenib (Fig. 4g), and the remaining ten trajectories appear in SI Fig. S3.

Hydrogen bond network persistence

We evaluated hydrogen bond occupancy across the full trajectory using Baker–Hubbard geometric criteria [11] as implemented in MDTraj [12] (Fig. 2).

Across three independent M327I–Revumenib replicates, the Trp346(NE1) occupancy averages 52±47% over the 100–500 ns window (individual replicates: 67%, 0%, 90%). Three independent WT–Ziftomenib replicates yield 93±1% (individual replicates: 94%, 92%, 94%), interrupted only by brief thermal excursions (Fig. 2a). The large inter-replicate variance in M327I–Revumenib reflects two distinct modes of instability: two replicates formed the Trp346 anchor contact within the first 50 ns and subsequently showed intermittent disengagement, while one replicate never achieved stable anchor contact despite the ligand remaining within the binding pocket throughout (Trp346 occupancy 0%, Mut327 contact occupancy 71%). The time-resolved donor-acceptor

distance in 8E90 (Fig. 2b, green line) confirms this pattern, with Trp346(NE1) repeatedly exceeding the 3.5 Å threshold (Fig. 3a).

First, Tyr281 shows consistently close contact with Revumenib across all three M327I–Revumenib replicates (minimum distance 2.83±0.01 Å), consistent with M327I not uniformly disrupting polar anchoring; the perturbation appears selective for Trp346 while leaving Tyr281 engagement intact. Second, Asn287 forms a contact with Revumenib at 56±47% occupancy across three M327I–Revumenib replicates (individual: 85%, 1%, 81%), substantially above its level in any Ziftomenib replicate (0% in all six). This contact is ligand-specific, appears as short-lived transient segments, and co-occurs with Trp346 engagement where maintained. In the context of weakened Trp346 anchoring, this compensatory contact partially offsets dissociation pressure but does not fully restore equilibrium. By contrast, in WT–Ziftomenib, Tyr281 maintains near-unity contact (Fig. 2c, red line) as part of a multi-anchor polar network that also includes stable Trp346 engagement (93±1%, n=3), without requiring compensatory redistribution. Quantitative distance and occupancy metrics for all key pocket residues across the four systems are provided in Table S5.

Conformational sampling and correlated motions

Principal component analysis and cross-correlation mapping show no group-level distinctions at this sampling depth. Replicate pairs drawn from one complex are no more similar to each other than pairs drawn from two different complexes sharing the same inhibitor and the same construct. For the correlation maps, the mean similarity of long-range correlation patterns is 0.70±0.05 within a complex against 0.69±0.05 between complexes in the dimeric systems, and 0.75±0.07 against 0.74±0.03 in the monomeric systems. For the principal component subspaces, the root mean square inner product over the first ten eigenvectors is 0.52±0.03 within a complex against 0.51±0.07 between complexes in the dimeric systems, and 0.52±0.07 against 0.51±0.06 in the monomeric systems. A within-complex inner product near 0.5 indicates that the essential subspace is itself not reproducible between replicates of one system, so the variance fraction carried by any individual principal component is not a stable descriptor at this trajectory length. Full distributions are given in Fig. S6, between the two inhibitors or between the two protein backgrounds. An earlier analysis of one trajectory per complex described attenuated and fragmented coupling in the M327I–Revumenib system relative to WT–Ziftomenib. The difference between those two correlation maps is 0.13, against 0.13±0.03 for pairs of replicates of the same complex, so that description is not separable from replicate-to-replicate variation.

Binding free energy and per-residue energy decomposition

MM/GBSA free energy calculations [13, 14] and per-residue energy decomposition over the 100–500 ns window provide effective enthalpic estimates (Fig. 5; Table S4). The MM/GBSA protocol employed here does not include entropic contributions and provides an effective enthalpic estimate rather than a true binding free energy.

Mean binding energies are comparable across complexes, but M327I–Revumenib shows larger between-replicate variance (3.4 kcal/mol) than WT–Ziftomenib (1.0 kcal/mol) (Fig. 5a and Table S4).

The per-residue decomposition for M327I–Revumenib (Fig. 5b) should be interpreted qualitatively. With n=3 replicates per condition, Ile327 appears among the contributors at approximately −5 kcal/mol (replicate mean; SD 3.97 kcal/mol), consistent with forced short-range contact arising from the steric perturbation of the pocket geometry. The compensatory Asn287 hydrogen bond (Fig. 2a) and the elevated between-replicate variability of the total binding energy support the interpretation that the M327I substitution destabilizes rather than enhances Revumenib binding.

The per-residue profile of WT–Ziftomenib (Fig. 5c) shows a different organizational pattern, with binding energy distributed across at least fifteen residues. The two largest individual contributors are Glu368 and Arg335 (approximately −12 and −11 kcal/mol, respectively; qualitative estimates), both reflecting electrostatic contacts that are absent from the Revumenib pharmacophore. Most relevant to the resistance question, Met327 ranks among the lower contributors at approximately −4 kcal/mol. This limited sensitivity to position 327 is consistent with the geometric observation that the extended scaffold of Ziftomenib engages the pocket through a distributed network involving Trp346, Tyr281, Tyr328, Tyr324, and Met283 that largely bypasses position 327. If wild-type Met327 already contributes only modestly, the M→I substitution would not be expected to substantially alter the total binding thermodynamics, consistent with the asymmetric clinical resistance data reported by Bourgeois and colleagues [6]. The Arg335 electrostatic subsite, which Revumenib does not engage, identifies a potentially underexploited pharmacological region that may warrant consideration in next-generation inhibitor design.

Discussion and conclusion

Our simulations provide a time-resolved perspective on M327I-mediated resistance. A prior crystallographic study [5] proposed that the branched Ile327 side chain generates steric conflict with the Revumenib cyclohexyl ring and

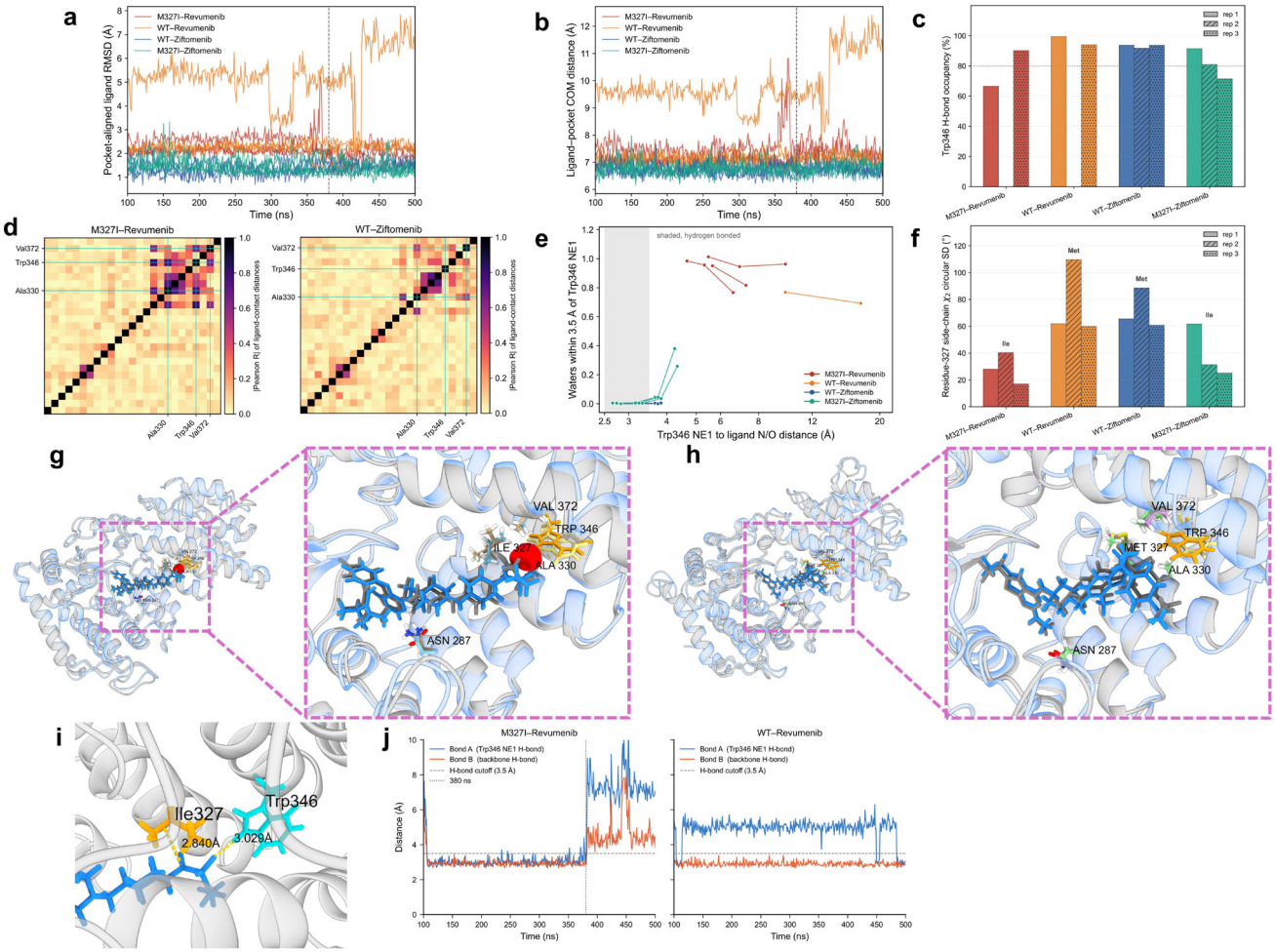


Fig. 4 Ligand stability and binding architecture analysis. a Pocket-aligned ligand RMSD time series across the four systems with three replicates each. One WT–Revumenib replicate shows substantial reorientation (pre-post RMSD 4.96 Å). **b** Ligand–pocket center-of-mass distance time series. **c** Trp346 hydrogen-bond occupancy per replicate. **d** Contact-distance correlation heatmaps for M327I–Revumenib and WT–Ziftomenib. Cyan lines mark clinical positions 346, 330 and 372. **e** Mean number of water oxygens within 3.5 Å of Trp346 NE1 as a function of the NE1 to ligand minimum distance at 0.1 ns sampling, one marker per trajectory rather than a group mean. A point is drawn only where that trajectory spends at least 100 frames (10 ns) in the interval, so an absent point means the interval was not sampled by that replicate. Counts are zero throughout the engaged range in all ten trajectories that sample it, the remaining two having lost the ligand before the analysis window opens, and saturate near one water beyond 4.5 Å. The two inhibitors occupy disjoint intervals at this frame threshold, the highest Ziftomenib interval being 4.0–4.5 Å and the lowest Revumenib interval 4.5–5.0 Å, so the profiles cannot be compared at matched distance. **f** Residue-327 side-chain χ_2 circular standard deviation. **g** Representative structures from clustering analysis at the 380 ns boundary (largest-cluster centroids, pre/post overlay) for M327I–Revumenib and WT–Ziftomenib (replicate 1 of each), shown as global and local views with water molecules near Trp346 NE1 rendered as red spheres. **h** The same two systems (M327I–Revumenib and WT–Ziftomenib) without water molecules, showing the pre/post ligand and pocket overlay directly. Representative structures for the remaining ten trajectories appear in SI Fig. S3. **i** Structural snapshot of M327I–Revumenib replicate 1 (pre-380 ns cluster centroid) rendered in ChimeraX, showing the two sulfonamide hydrogen bonds that bridge the helix-turn-helix module. Bond A: Trp346 NE1 (indole nitrogen) to sulfonamide O3x, 3.03 Å. Bond B: sulfonamide N6x to the Ile327 backbone carbonyl oxygen, 2.84 Å. Gold dashed lines indicate each hydrogen bond; distances are labeled. **j** Time series of Bond A (dark blue) and Bond B (orange) donor–acceptor distances for M327I–Revumenib replicate 1 (left panel) and WT–Revumenib replicate 1 (right panel), 100–500 ns. The vertical dashed line at 380 ns marks the onset of anchor disengagement in M327I–Revumenib

disrupts the Trp346 hydrogen bond. Our time-resolved trajectories suggest this is not the primary determinant. Ile327 and Met327 sample comparable distance from and orientation toward the ligand cyclohexyl ring (Fig. 3), not consistent with a simple steric displacement mechanism. Instead, the simulations suggest an intrinsically fragile Trp346 anchor whose disengagement is not specific to M327I, as one of three WT–Revumenib replicates also lost the anchor (Fig. 2). Within this framework, M327I side-chain rigidity appears to exacerbate an existing fragility rather than create it, though these remain hypotheses requiring experimental validation. The trajectories add temporal resolution by revealing a metastable

phase in which residual anchoring is temporarily maintained before the Trp346 anchor weakens beyond recovery (Figs. 3 and 2). This interpretation is broadly consistent with the agent-selective resistance patterns documented in base-editor screening [6]. The high Ile327 energy contribution (Fig. 5) reflects forced short-range contact imposed by the pocket geometry, and the broader cost of this arrangement is apparent in impaired Trp346 engagement, compensatory Asn287 hydrogen bonding, and elevated ΔG_{bind} variance. On the Ziftomenib side, the minimal energetic contribution of Met327 reflects the geometric architecture in which the extended scaffold of Ziftomenib engages the pocket through a decoupled distributed hydrophobic network that largely bypasses position 327; this structural feature may contribute to the differential resistance landscape between agents [6]. A comparative summary of the key structural, dynamic, and thermodynamic features that distinguish the two binding architectures is provided in Table 1.

A single wild-type Ziftomenib trajectory shows a pronounced fluctuation near residue 114 that is absent from the other two replicates of the same system and from all other trajectories, so it is not treated as a system-level feature here. We therefore do not propose a link between inhibitor binding and the Menin surfaces engaged by co-regulatory partners. Whether either inhibitor perturbs those interfaces is a question that correlated-motion analysis of this length cannot address, and would be better approached by hydrogen-deuterium exchange mass spectrometry [15] or nuclear magnetic resonance relaxation measurements [16]. From a drug design perspective, the observations reported here suggest three potential design considerations, though these proposals require experimental validation through structure-activity relationship studies and binding affinity measurements. First, inhibitors whose binding efficacy depends on a single contact residue may be vulnerable to substitutions at or adjacent to that residue. Next-generation Menin inhibitors might therefore benefit from distributing binding energy across a geometrically redundant contact network that reduces the impact of any single-point mutation, as the Ziftomenib binding mode in the wild-type pocket appears to illustrate [3, 17]. Second, the large Arg335 electrostatic contribution in the 8VA6 complex, approximately −11 kcal/mol, arising from a polar contact that the Revumenib pharmacophore does not engage (Fig. 5), identifies a potentially underexploited subsite that may warrant consideration in next-generation inhibitor design. Introducing functional groups such as carboxylate, phosphonate, or sulfonamide moieties capable of engaging Arg335 might provide an electrostatic anchor that is geometrically independent of position 327 and therefore potentially less susceptible to the steric perturbation characterized here. Such anionic moieties have been systematically employed in structure-based drug design to exploit binding-site arginines through bidentate electrostatic contacts, with guanidinium-anion salt bridges typically contributing 3 to 8 kcal/mol of binding free energy depending on geometry and solvent exposure [18, 19]. However, introduction of ionizable groups must be balanced against potential impacts on membrane permeability, oral bioavailability, and pharmacokinetic properties, which require experimental structure–activity relationship studies to assess. Third, the capacity to accommodate a favorable protein conformational rearrangement may itself be a pharmacologically relevant property [20], as the ligand-specific conformational adaptation associated with Ziftomenib binding may provide a buffer against pocket perturbation by allowing the protein to accommodate moderate geometric changes while maintaining binding contacts (Fig. 4g, h).

At 500 ns per replicate, rare conformational transitions may remain incompletely sampled, and infrequent events observed within individual replicates may disproportionately influence the reported ensemble averages. The Trp346 NE1 donor–acceptor distance in the Revumenib systems begins at 7.03–7.82 Å across six replicates, reflecting the 8E90 crystal geometry in which no Trp346 hydrogen bond is present, and converges below 3.5 Å in four of six replicates within 15–51 ns of production. The early window therefore represents equilibration from the crystal starting geometry rather than a mutation-specific relaxation, since both the wild-type and M327I backgrounds start at comparable distances and form the anchor contact in the same time frame. Similarly, although equilibrium MD propagated from crystal structures cannot unambiguously distinguish ligand-associated conformational adaptation from passive crystallographic relaxation, this limitation is partially mitigated here because the two source structures differ primarily in the bound inhibitor: the binding-pocket Cα RMSD between 8E90 and 8VA6 is 0.19 Å under pocket-own superposition and 0.23 Å after global backbone alignment, so the early-trajectory differences between the two inhibitor systems reflect inhibitor identity rather than starting-structure divergence. The use of conventional unbiased molecular dynamics without enhanced sampling methods such as replica-exchange molecular dynamics [21] or metadynamics [22] may nonetheless leave rare conformational states incompletely sampled. Applying such methods in future work would help determine whether the anchor weakening observed beyond 380 ns represents a thermodynamically favorable transition or a rare but potentially reversible excursion. A further caveat is that the anchor-loss events, although reproducible across two of three M327I–Revumenib replicates and one of three WT–Revumenib replicates, are in silico observations whose correspondence to experimentally observable binding modes remains to be established, and orthogonal validation by time-resolved biophysical measurements such as NMR relaxation dispersion [16] or hydrogen-deuterium exchange mass spectrometry [15]

Table 1 Summary of key structural, dynamic, and thermodynamic features of the two Menin–inhibitor complexes. All residue numbers follow the clinical 615-amino acid isoform convention. SD: standard deviation

Feature	8E90: Menin(M327I)–Revumenib	8VA6: Menin(WT)–Ziftomenib
Construct and source crystal	Homodimer (chains A+B), PDB 8E90, 1.85 Å	Monomer (chain A), PDB 8VA6, 1.57 Å
Protein state	M327I drug-resistant mutant	Wild-type
Residue at position 327	Ile (mutant)	Met (wild-type)
Protein backbone RMSD	2.15±0.28 Å (mean±SD, n=3, 100–500 ns)	2.47±0.26 Å (mean±SD, n=3)
Radius of gyration	25.10±0.13 Å (mean±SD, n=3)	25.16±0.08 Å (mean±SD, n=3); all four complexes fall within 0.19 Å
Trp346 distance stability	Anchor disengaged in 2 of 3 replicates near 380 ns; also lost in 1 of 3 WT–Revumenib replicates	Stable <3.5 Å throughout 500 ns
Effect of residue 327 on ligand	Ile327 and Met327 sample comparable ligand proximity; no distinct steric displacement	Met327 lowest contributor; minimal steric effect
PCA PC1 variance explained	27.5±12.2% (n=3)	29.7±5.4% (n=3); ranges overlap across all four complexes
DCCM character	Mean long-range RMS difference 0.144±0.044 (n=3, $\|i - j\| > 50$)	0.151±0.016 (n=3); no correlation feature distinguishes the four complexes
ΔG_{bind} stability (SD)	Larger SD (elevated binding variability)	Smaller SD (consistent binding)
Energy contribution at position 327	Approximately −5 kcal/mol (likely forced steric contact; ninth of 462 residues, behind Tyr281, Met283 and Asn287)	Approximately −4 kcal/mol (limited sensitivity; lowest contributor)
Mutation tolerance	Lower (single-site substitution coincides with anchor weakening)	Higher (position 327 not a major network node)

would provide independent evidence. Absolute free-energy predictions for the mutant-induced potency shifts were likewise not attempted, so the simulations are intended to inform mechanistic hypotheses rather than to quantitatively predict the magnitude of clinical resistance. Additionally, the principal component analysis, dynamic cross-correlation, and MM/GBSA analyses presented here are complementary descriptors derived from the same molecular dynamics trajectories. Their mutual consistency strengthens the internal coherence of the proposed model but does not constitute multiple independent experimental validations, as all analyses share the same underlying force field, sampling protocol, and trajectory ensemble.

The simulations suggest that asymmetric M327I resistance may arise from contrasting binding architectures. The identification of Arg335 as a potentially underexploited electrostatic subsite may inform next-generation inhibitor design, though such recommendations require experimental validation.

Materials and methods

System preparation and residue numbering convention

Two co-crystal structures were retrieved from the Protein Data Bank: entry 8E90 (Menin(M327I)–Revumenib, resolution 1.85 Å) and entry 8VA6 (Menin(WT)–Ziftomenib, resolution 1.57 Å). Missing heavy atoms within crystallographically resolved regions were identified and added using PDBFixer [23]; missing residues (terminal and loop segments absent from the crystal structures) were not modeled. Both structures were complete in the binding-site region. After preparation, each chain contained 462 residues (clinical numbering spans positions 3 to 610, with engineered deletions accounting for the deficit relative to the 615-residue isoform). All crystallographic water molecules, including those in the binding site, and all other heteroatoms were removed. The ligands Revumenib (PDB residue code OQ4) and Ziftomenib (PDB residue code K5O) were extracted from the respective co-crystal coordinates using RDKit [24] prior to system assembly.

All residue numbers in this study follow the clinical 615-amino acid isoform of Menin (UniProt O00255). The MD topologies renumber protein residues contiguously from one, so the correspondence to clinical numbering is piecewise rather than a single offset. Each chain contains three crystallographic gaps, giving four segments with distinct offsets: 6, 19, 32, and 126 for the Revumenib systems and 6, 26, 38, and 125 for the Ziftomenib systems. All pocket residues listed in Table 2 fall inside the second segment, where the offsets are 19 and 26, respectively. Residue-level quantities reported over the whole chain use the full piecewise mapping, reconstructed by ordered alignment between each MD topology and its source crystal structure. One consequence

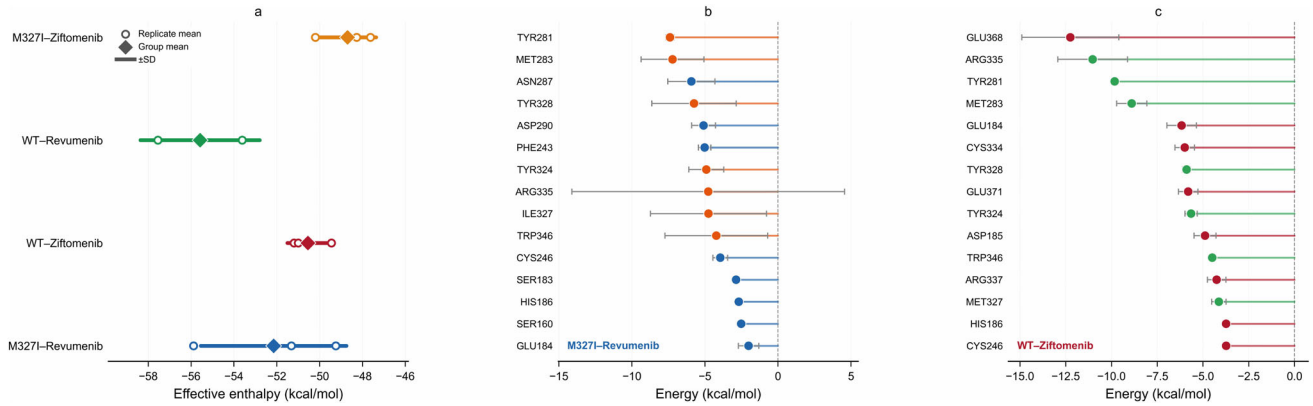


Fig. 5 Binding free energy and per-residue energy decomposition across all four Menin–inhibitor complexes. a Effective enthalpy (ΔG_{bind}, kcal/mol) for all four complexes estimated by single-trajectory MM/GBSA over the 100–500 ns window. Diamonds mark group means; open circles show individual replicate means; horizontal bars show replicate standard deviation ($N = 3$, or $N = 2$ for WT–Revumenib; full statistics in Table S4). **b** Per-residue energy decomposition for M327I–Revumenib (8E90), top 15 residues by group mean. Error bars represent replicate standard deviation. **c** Per-residue energy decomposition for WT–Ziftomenib (8VA6), top 15 residues. Per-residue values are qualitative indicators of relative contributions; they depend on the GB model, exclude entropic terms, and do not represent physical interaction energies [14]

is that clinical residue 470 lies inside a gap spanning clinical 463 to 556 and is absent from the simulated construct.

Comparison of starting structures

The two source structures share 452 protein residues in common; residue names match at 450 of those positions (99.6%, denominator: co-observed positions only, excluding regions absent from one structure). The two exceptions are both engineered substitutions: T5A at the expression-tag boundary and I322M, which corresponds to the M327I position of interest (crystal numbering). After optimal superposition of the 452 common Cα positions by Kabsch rotation, the overall backbone RMSD is 1.24 Å, driven principally by differences at the boundaries of the respective engineered deletions. The seven binding-site residues (Trp346, Tyr281, Tyr324, Met/Ile327, Met283, Tyr328, Asn287 in clinical numbering) deviate by only 0.23 Å from this global alignment and 0.19 Å under their own superposition, indicating that the starting pocket geometry is equivalent between the two systems at the relevant precision. The two crystal structures differ in chain composition (8E90: homodimer, chains A and B; 8VA6: monomer, chain A), resolution (1.85 Å versus 1.57 Å), construct design, and solvated system size (219,095 versus 109,836 atoms in 131.4 Å versus 104.7 Å cubic boxes). The crystallographic gap patterns also differ: the 8E90 construct has two genuine unmodelled chain breaks (residues 53–65 and 388–400 in crystal numbering, verified by Cα-to-Cα distances of 28.5 Å and 11.7 Å across the junctions), while the equivalent regions in 8VA6 are engineering deletions with intact main-chain geometry (Cα-to-Cα 3.8 Å). These global construct differences mean that cross-inhibitor comparisons of whole-chain dynamics would carry a structural confound; all interpretive comparisons in this study are therefore made within inhibitor pairs, in which the only variable is the residue at position 327.

Construction of reciprocal complexes

The two reciprocal complexes were constructed by in silico point mutation using PDBFixer. For the WT–Revumenib system, the 8E90 co-crystal carries the I322M engineering substitution (I327 in clinical numbering); the I322M reversion to Met was applied to both chains A and B of the cleaned 8E90 structure. For the M327I–Ziftomenib system, the substitution ILE301ILE was applied to chain A of the 8VA6 structure. In both cases, the missing-residues dictionary was set to empty to preserve the parent sequence at all other positions; only the mutated side-chain heavy atoms were rebuilt. Side-chain geometry was not optimized at this stage; rotamer adjustment was carried out implicitly during the energy minimisation described below.

Ligand parameterization

Canonical SMILES strings for Revumenib and Ziftomenib were used as chemical templates to establish correct bond orders, formal charges, and tautomeric states. Both ligands are neutral at physiological pH. For each ligand, an initial three-dimensional conformer was generated from the SMILES string using RDKit, and bond orders were subsequently assigned by maximum common substructure (MCS) alignment between the SMILES-derived molecule and the crystallographic geometry, using atom-type-agnostic and bond-type-agnostic comparison modes. Heavy-atom coordinates from the crystal structure were transferred to the

corrected molecular graph, preserving experimental binding geometry while establishing chemically valid connectivity. Quality control confirmed that the root mean square deviation between the original and bond-order-corrected heavy-atom positions was below 0.05 Å for both ligands and that both corrected structures passed standard chemical validation without error.

Partial atomic charges were assigned by the AM1-BCC method [25] through the OpenFF Toolkit [26], and ligand force-field parameters were generated with the OpenFF Sage 2.0.0 small-molecule force field [27] using the OpenMM force field generation interface [23].

Molecular dynamics simulation protocol

Protein topology and force-field parameters were assigned with the AMBER ff14SB force field [28]. All existing hydrogen atoms from PDBFixer output were removed and replaced by hydrogen atoms added at pH 7.4 using the OpenMM structure preparation interface to ensure self-consistent protonation states under the AMBER14 force field. Histidine tautomeric states were assigned automatically by the force field based on the local hydrogen-bonding environment; the resulting assignments were predominantly HID (Nδ1-protonated) with HIE (Nϵ2-protonated) at two positions per chain (residues 167 and 303 in the 8E90 numbering, corresponding to clinical positions 186 and 322; residues 160 and 296 in the 8VA6 numbering, corresponding to clinical positions 186 and 322). No histidine residues were doubly protonated (HIP). Each protein–ligand complex was solvated in a rectangular TIP3P water box [29] with a minimum solute-to-boundary padding of 1.0 nm. Sodium and chloride counter-ions were added to neutralize the net charge and achieve a physiological ionic strength of 0.15 M. All simulations were executed with OpenMM [23] on a single NVIDIA GeForce RTX 4090 GPU using the CUDA platform with mixed-precision arithmetic.

System preparation and equilibration followed three sequential stages. Energy minimisation was performed with the L-BFGS algorithm until the maximum atomic force fell below 10.0 kJ mol^{-1} nm^{-1} with no upper limit on iteration count. NVT heating was carried out for 100 ps at 300 K using a Langevin integrator with a friction coefficient of 1.0 ps^{-1} and a 2 fs time step. NPT density equilibration followed for an additional 100 ps under the same thermostat settings with a Monte Carlo barostat targeting 1.0 bar, with trial volume moves attempted every 25 integration steps. Throughout both equilibration stages, harmonic position restraints with a force constant of 1000 kJ mol^{-1} nm^{-2} were applied to all non-hydrogen protein and ligand atoms via a custom external force using a periodic distance potential; water molecules and monovalent ions were left unrestrained.

Production simulations of 500 ns were then conducted under NPT conditions using the same Langevin integrator and Monte Carlo barostat, with all positional restraints removed. Atomic coordinates were saved every 100 ps (50,000 integration steps), yielding 5000 frames per trajectory. Long-range electrostatic interactions were treated with the Particle Mesh Ewald method [30, 31] throughout all stages, with a real-space cutoff of 1.0 nm and an Ewald error tolerance of 5×10^{-4}.

Trajectory analysis

All post-processing was performed in Python using MDTraj [12]. Periodic boundary artefacts were first corrected by imaging molecules into the primary unit cell. Each trajectory was then sliced to retain protein and ligand atoms only. For structural and conformational analyses (global metrics, hydrogen bond profiling, and principal component analysis), the full 0–500 ns trajectory was used, with frames subsampled at a stride of 10 relative to the 100-ps saving interval, yielding an effective analysis interval of 1 ns and 500 analysis frames per system. For binding free energy calculations and the convergence assessment, the first 100 ns were treated as equilibration and excluded, and frames were analyzed over the 100–500 ns window; for MM/GBSA, a 1 ns sampling interval was applied, yielding approximately 400 frames per system. Per-residue fluctuation, principal component analysis, cross-correlation, and anchor-site hydration were recomputed at the full 100 ps saving interval over the 100–500 ns window, giving 4000 frames per trajectory, because the 1 ns interval proved too coarse to resolve anchor-site water exchange.

Table 2 Residue numbering correspondences between the clinical 615-aa isoform, the original PDB chain (610-aa, UniProt O00255), and the MD sequence indices used in this study

Role	Clinical	PDB	8E90-MD	8VA6-MD
M327I resistance site	327	322	308	301
Trp346 anchor	346	341	327	320
Thr349 resistance site	349	344	330	323
Tyr281 (pocket)	281	276	262	255
Tyr324 (pocket)	324	319	305	298
Tyr328 (pocket)	328	323	309	302
Met283 (pocket)	283	278	264	257

Global structural metrics were computed after superposing all frames onto the first production frame using backbone heavy atoms. Protein backbone root mean square deviation (RMSD) and per-residue Cα root mean square fluctuation (RMSF) were calculated with standard trajectory analysis methods. The radius of gyration (Rg) was computed on protein atoms alone using standard geometric analysis. Ligand RMSD was evaluated on ligand heavy atoms using the same superposition reference. All distance and displacement values are reported in angstroms. RMSF residue indices were converted to clinical 615-aa numbering by applying the per-system offsets defined in "System preparation and residue numbering convention."

Convergence and equilibration were assessed for all 12 trajectories (four complexes × three independent replicates). For this assessment, the single-chain Cα RMSD of the ligand-binding chain was used, superposing each frame on the Cα atoms of that chain alone; for the dimeric systems (M327I–revumenib, WT–revumenib), this single-chain treatment avoids an inter-chain periodic-boundary artifact that otherwise inflates the apparent RMSD. Equilibration was judged from the standard deviation of the backbone RMSD over the 100–500 ns window, the stationarity of its mean across consecutive 50 ns blocks, and the stability of the radius of gyration. The effective number of independent samples (N_{eff}) per trajectory was estimated by Flyvbjerg–Petersen block averaging [32] of the backbone RMSD over the 100–500 ns window. Convergence metrics for all trajectories are reported in Figs. S1 and S2 and Table S1.

Minimum distances between ligand heavy atoms and the sidechain heavy atoms of each key pocket residue were computed frame-by-frame over all pairwise combinations, retaining the minimum value across all pairs for each frame. Hydrogen atoms were excluded from both the ligand and protein side chains in these calculations.

Hydrogen bond occupancy was evaluated using Baker–Hubbard geometric criteria: donor-to-acceptor distance below 3.5 Å and donor-H$\cdots$acceptor angle above 120°, applied to all potential donor-acceptor pairs. Per-residue occupancy was defined as the fraction of trajectory frames in which a hydrogen bond between the ligand and the given protein residue simultaneously satisfied both criteria.

To assess whether π–π stacking contributes to the Trp346 interaction, the Trp346 indole six-membered ring (atoms CD2, CE2, CE3, CZ2, CZ3, and CH2) and all aromatic rings of each ligand were identified by ring perception and planarity testing (singular value decomposition residual below 0.10 Å). The centroid-to-centroid distance and the inter-ring plane angle were computed per frame using the minimum-image convention. π–π stacking occupancy was defined as the fraction of frames satisfying: centroid distance below 5.5 Å and either a parallel geometry (interplane angle below 30° with lateral offset below 2.0 Å) or a T-shaped geometry (interplane angle above 60°). The hydrogen bond and π–π stacking criteria use distinct atomic subsets (NE1 donor atom versus aromatic ring centroids) and are therefore independent. Across all 12 trajectories, the π–π stacking occupancy was 0% for every replicate; the minimum centroid distance from the Trp346 ring to the nearest Revumenib aromatic ring was approximately 18 Å (range 17.5–19.1 Å) and to the nearest Ziftomenib ring approximately 7.3–8.0 Å, both well beyond the stacking criterion. Three additional interaction classes were screened using the following geometric criteria: C–H$\cdots\pi$ contact (ligand carbon to indole face centroid $\leq$4.5 Å, off-normal angle $\leq$40°); π-cation interaction (ligand nitrogen to indole face centroid $\leq$6.0 Å, off-normal angle $\leq$40°); and halogen bond (ligand fluorine to any indole atom $\leq$3.5 Å). All three classes showed 0% occupancy in every replicate (Table S2). The Trp346–ligand interaction is therefore attributed solely to the NE1 hydrogen bond.

Principal component analysis (PCA) [33, 34] and cross-correlation analysis [35] were performed independently for each of the 12 trajectories on chain-0 Cα atoms over the 100–500 ns window, after iterative least-squares superposition onto the window-mean structure. Cross-correlation matrices were computed as $\mathrm{DCCM}_{ij} = \langle \Delta r_i \cdot \Delta r_j \rangle / \sqrt{\langle \Delta r_i^2 \rangle \langle \Delta r_j^2 \rangle}$ and averaged within each group of three replicates on the 453-residue axis common to all 12 topologies.

Whether a descriptor of global dynamics separates the four complexes at all was tested before any group comparison was reported. Replicate pairs were split into within-complex pairs (same complex, different replicate, $n = 6$ per topology) and between-complex pairs (same inhibitor and same topology, different protein background, $n = 9$ per topology), and the two similarity distributions were compared. Pairs were restricted to one topology because the revumenib systems are dimeric and the ziftomenib systems are monomeric. Similarity was quantified as the Pearson correlation and the root mean square difference between correlation matrices, taken over all residue pairs and over long-range pairs ($|i - j| > 50$) separately, and as the root mean square difference between per-residue fluctuation profiles. Profile comparisons exclude three residues at each end of every contiguous segment, retaining 429 of 453 positions, since chain and segment termini carry boundary fluctuation that would otherwise dominate the comparison. Overlap of the essential subspaces was quantified as the root mean square inner product over the first ten eigenvectors. Where the within-complex and between-complex distributions coincide, the descriptor carries no group-specific signal at this sampling depth, and no group comparison based on it is interpreted.

Pocket-aligned ligand RMSD

To isolate ligand motion relative to the binding pocket independent of global protein rotation, each trajectory frame was

superposed to the first frame using binding-site Cα atoms (clinical positions 281, 283, 287, 324, 327, 328, and 346, corresponding to Tyr281, Met283, Asn287, Tyr324, residue 327, Tyr328, and Trp346). After alignment, ligand heavy-atom RMSD was computed without further superposition. This procedure separates local ligand displacement within the pocket from whole-protein rigid-body motion.

Ligand–residue contact correlation

For each protein residue, the minimum heavy-atom distance to any ligand heavy atom was computed per frame. Residues with contact occupancy at or above 20% (distance at or below 4.5 Å) were retained. Pearson correlation coefficients were computed between all pairs of residue distance time series across the equilibrated window (100–500 ns). Background correlation was estimated from pairs of non-adjacent residues (sequence separation above 10). Coupled modules were identified as residue groups exhibiting an absolute correlation substantially above background.

Side-chain conformational flexibility

Residue-327 side-chain χ_2 dihedral angles (N–Cα–Cβ–Cγ for Met, N–Cα–Cβ–Cγ1 for Ile) were extracted per frame. Conformational flexibility was quantified as the circular standard deviation across the equilibrated trajectory window, computed using standard circular statistics to account for the 2π periodicity of dihedral angles.

Clustering and representative structure selection

Frames before (100–380 ns) and after (380–500 ns) the anchor-loss transition were clustered independently using hierarchical clustering (average linkage) with a distance cutoff of 2.5 Å RMSD. Clustering features comprised pocket Cα atoms plus ligand heavy atoms, computed after superposing all frames to the first frame using pocket Cα atoms. The representative structure for each window was defined as the centroid of the largest cluster (the frame with minimum mean RMSD to other cluster members).

Pocket water quantification

Water oxygens within 3.5 Å of Trp346 NE1 (indole nitrogen) were counted per frame at the full 100 ps saving interval. The cutoff captures first-shell solvation while excluding bulk solvent. Counts are reported as a function of the NE1 to ligand minimum distance rather than of time, and per trajectory rather than pooled across replicates, because a bin pooled over replicates can be carried almost entirely by a single trajectory. A bin is reported for a trajectory only where that trajectory spends at least 100 frames (10 ns) in it.

Anchor disengagement events were defined by a two-threshold criterion: rupture when the NE1 to ligand distance exceeds 5.0 Å and recovery when it returns below 3.5 Å, with the state held between the two thresholds. A rupture was counted as sustained when the disengaged state persisted for at least 2 ns, or to the end of the trajectory. A single threshold placed at the 3.5 Å hydrogen bond criterion counts every fluctuation across that value as a separate event, giving a median dwell time of one frame and assigning the largest event counts to the trajectories whose anchor distance happens to sit nearest the threshold. Under the two-threshold criterion, the 12 trajectories contain 24 ruptures, of which four are sustained.

Binding free energy calculations

Binding free energies and per-residue energy contributions were estimated by a single-trajectory MM/GBSA protocol implemented in Python using OpenMM [23] and MDTraj [12]. Frames were sampled at 1 ns intervals from the 100–500 ns segment of each production trajectory (400 to 401 frames per system), discarding the initial 100 ns to exclude residual equilibration effects. Sampling at 2 ns was run as a sensitivity check and is reported in Table S4.

Molecular mechanics energies comprising van der Waals (ΔE_{vdW}) and electrostatic (ΔE_{elec}) contributions were computed as vectorized pairwise Lennard–Jones and Coulomb interactions between all protein–ligand atom pairs, using nonbonded parameters extracted from the OpenMM serialized system files. All pairwise interactions were evaluated without a distance cutoff.

Polar solvation free energy ($\Delta G_{\mathrm{polar}}$) was estimated using the GB-OBC2 implicit solvent model [36], implemented through persistent OpenMM simulation contexts constructed separately for the protein–ligand complex, isolated receptor, and isolated ligand at each sampled frame. Standard mbondi2 atomic radii were applied, with solute and solvent dielectric constants of 1.0 and 78.5, respectively, and the non-bonded method set to no cutoff.

Non-polar solvation free energy of each species was modeled as $G_{\mathrm{np}} = \gamma \cdot \mathrm{SASA} + \beta$, where $\gamma = 0.00542\ \mathrm{kcal\ mol^{-1}\ Å^{-2}}$ and $\beta = 0.92\ \mathrm{kcal\ mol^{-1}}$. For binding, the per-species offsets cancel ($\beta_{\mathrm{cpx}} - \beta_{\mathrm{rec}} - \beta_{\mathrm{lig}} = -\beta$), giving the binding contribution $\Delta G_{\mathrm{np}} = \gamma \cdot \Delta\mathrm{SASA} - \beta$. Solvent-accessible surface area was computed using the Shrake–Rupley rolling-sphere algorithm [37] as implemented in MDTraj, applied independently to the complex, receptor-only, and ligand-only atom-slice sub-trajectories to obtain the burial area ΔSASA.

Per-residue energy decomposition was performed by partitioning the molecular mechanics pairwise interaction energies ($\Delta E_{\mathrm{vdW}} + \Delta E_{\mathrm{elec}}$) by protein residue; polar and non-polar solvation terms were not decomposed at the per-residue level. Entropic contributions were not included in this

analysis; the reported ΔG_{bind} values therefore represent an effective enthalpic estimate, and absolute magnitudes should not be directly compared with experimental binding affinities. The mechanistic conclusions rest primarily on relative energetic differences and per-residue decomposition patterns between the two complexes [14].

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00894-026-07014-w.

Author contribution Y.W. conceived and designed the study, developed the methodology and software, performed the simulations and formal analysis, curated the data, prepared all figures, wrote the original draft, revised the manuscript, and reviewed and edited the manuscript. K.W. contributed to the methodology, formal analysis, data curation, manuscript revision, project supervision, and funding acquisition.

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Data availability Molecular dynamics trajectories (XTC format) are available at Zenodo: https://doi.org/10.5281/zenodo.21853260. Analysis scripts are available at GitHub: https://github.com/wyqmath/menin-m327i-md-analysis.

Declarations

Conflict of interest The authors declare no competing interests.

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